

# DOMAIN COPILOT

## An Agentic RAG Platform

Assessment-Aligned Agile Backlog, UI Wireframe Specification & 12-Day Engineering Roadmap

|                 |                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------|
| ASSIGNED DOMAIN | D<n> - MUST be replaced from invitation email before implementation/submission                       |
| MANDATORY TWIST | T<n> - MUST be replaced from invitation email before implementation/submission                       |
| STACK           | Node.js / TypeScript + React / Next.js + Tailwind + Shadcn UI + PostgreSQL/pgvector + Docker Compose |

**Variant safety gate:** The ITI brief says the invitation email determines both Domain and Twist. If the email does not state them, derive Domain = (last two National ID digits) mod 7 and Twist = (sum of all digits) mod 8. Building the wrong assigned variant invalidates the submission. This blueprint therefore treats D<n>/T<n> as locked placeholders, not as an assumption.

**Assessment basis:** This document is a planning artifact grounded in the supplied ITI assessment. It intentionally distinguishes the mandatory MVP floor (P0) from explicitly documented deferrals and treats documentation, teaching, security, evaluation and Git discipline as implementation work, not afterthoughts.

## PART 1 - Visual Analysis & UI Language

The supplied reference screenshot uses a compact technical backlog grammar: a deep navy epic band, steel-gray story bands, pale-blue technical blocks, green acceptance blocks, restrained typography, and status/priority cues. The platform should borrow this visual grammar without spending MVP hours on decorative design.

| Reference cue    | Treatment                                                                         | Domain Copilot mapping                                          |
|------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Epic banner      | Deep navy; white title; compact story-count indicator                             | Feature/page header and section title                           |
| Story band       | Steel gray; white ID/title; compact SP/Priority chips                             | Task cards, agent runs, approval items                          |
| Technical panel  | Pale blue                                                                         | API contracts, architecture constraints, trace metadata         |
| Acceptance panel | Soft green                                                                        | Passed criteria, grounded evidence, completed pipeline stages   |
| Risk / warning   | Soft amber or soft red                                                            | Refusal state, failed ingestion, high-risk approval item        |
| Typography       | Arial/Inter-like sans; title > section > card > body                              | Dense enterprise content with readable hierarchy                |
| Navigation       | Fixed left sidebar, absent in source image so treated as an intentional extension | Dashboard, Corpus, Copilot, Reviews, Runs, Evaluation, Settings |

### Page layout system

|             |                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------|
| APP SHELL   | Sidebar 240px + top bar + route content; persistent role/session context.                                    |
| TOP BAR     | Navy strip, current domain/twist badge, provider badge, run status, profile.                                 |
| CARD RHYTHM | 8-12px outer gaps, 10-14px internal padding, gray card headers, evidence blocks.                             |
| STATUS CUES | P0 / deferred priority, queued/running/success/failed, grounded/refused, approval pending/approved/rejected. |
| DENSE DATA  | Tables for documents/runs/evaluation results; cards for stories, citations, trace spans.                     |

## PART 2 - Production-Grade Agile Backlog

Backlog convention: P0 = mandatory MVP floor for this assessment. Deferred work is documented in the gap table. Estimates are focused engineering points, not calendar days.

| EPIC 01                                                                                                  | Knowledge Ingestion & Vector Storage Pipeline | 8 stories |
|----------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----------|
| Extract, clean, chunk, embed, index; preserve source metadata, idempotency and visible ingestion health. |                                               |           |

| ING-001                  | Document upload & validation | 3 SP                                                                                                                                                                                                                                                                                                                          | P0 |
|--------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| USER STORY               |                              | As an enterprise expert, I want to upload supported documents so that they enter a controlled ingestion pipeline.                                                                                                                                                                                                             |    |
| TECHNICAL / API CONTRACT |                              | POST /api/documents multipart; validate MIME, extension, size, checksum; create document + ingestion job; reject unsupported/oversize input before persistence.                                                                                                                                                               |    |
| ACCEPTANCE CRITERIA      |                              | <ul style="list-style-type: none"> <li>✓ Allowed PDF/DOCX/TXT inputs are accepted.</li> <li>✓ Invalid MIME/content mismatch returns 400 with a descriptive error.</li> <li>✓ Upload response returns documentId, jobId and queued status.</li> <li>✓ No secret or raw file content is written to application logs.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                              | Frontend UI + Backend Core                                                                                                                                                                                                                                                                                                    |    |

| ING-002                  | Extraction & cleaning stages | 3 SP                                                                                                                                                                                                                                                                                                                                   | P0 |
|--------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| USER STORY               |                              | As an engineer, I want extraction and cleaning to be separate stages so that failures are diagnosable and testable.                                                                                                                                                                                                                    |    |
| TECHNICAL / API CONTRACT |                              | Implement Extract -> Clean interfaces returning typed stage results and failure codes; preserve source/page metadata.                                                                                                                                                                                                                  |    |
| ACCEPTANCE CRITERIA      |                              | <ul style="list-style-type: none"> <li>✓ Each stage is unit-testable in isolation.</li> <li>✓ Extraction failure persists a failed stage and actionable reason.</li> <li>✓ Cleaning removes boilerplate deterministically without losing source location.</li> <li>✓ Pipeline can resume or safely retry a failed document.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                              | Backend Core                                                                                                                                                                                                                                                                                                                           |    |

| ING-003                  | Structure-aware chunking | 3 SP                                                                                                                                                                                                                                                                                                                          | P0 |
|--------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| USER STORY               |                          | As an enterprise expert, I want chunks to preserve document structure so that citations remain meaningful.                                                                                                                                                                                                                    |    |
| TECHNICAL / API CONTRACT |                          | Document and justify a deterministic chunking strategy using headings, pages and domain structure; include stable chunk ids and offsets.                                                                                                                                                                                      |    |
| ACCEPTANCE CRITERIA      |                          | <ul style="list-style-type: none"> <li>✓ Same source/version yields stable chunk ids.</li> <li>✓ Chunk metadata includes source, section, page/clause and version where available.</li> <li>✓ No chunk crosses a citation boundary without explicit metadata.</li> <li>✓ Chunking decision is recorded in ADR-001.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                          | AI/Agent Engine                                                                                                                                                                                                                                                                                                               |    |

| ING-004                  | Embedding generation | 3 SP                                                                                                                                                        | P0 |
|--------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| USER STORY               |                      | As a system, I want embeddings generated through an abstraction so that the provider can be swapped without business-logic changes.                         |    |
| TECHNICAL / API CONTRACT |                      | Define embedding provider port; generate deterministic embeddings for validated chunks; validate dimension/model version before handoff to the index stage. |    |

|                     |                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACCEPTANCE CRITERIA | <ul style="list-style-type: none"> <li>✓ Embedding calls are hidden behind an interface.</li> <li>✓ Embedding dimension/model version are recorded with the chunk.</li> <li>✓ A provider failure produces a typed ingestion error.</li> <li>✓ Changing provider requires configuration plus one adapter.</li> <li>✓ Indexing does not run when embedding validation fails.</li> </ul> |
| ARCHITECTURAL LAYER | AI/Agent Engine + Database                                                                                                                                                                                                                                                                                                                                                            |

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| ING-005                  | Vector indexing                                                                                                                                                                                                                                                                                                                                       | 2 SP | P0 |
| USER STORY               | As a system, I want validated embeddings indexed separately so that vector search is reliable and index failures are diagnosable.                                                                                                                                                                                                                     |      |    |
| TECHNICAL / API CONTRACT | Implement a dedicated Index stage that writes validated embeddings to pgvector and builds/maintains the similarity index.                                                                                                                                                                                                                             |      |    |
| ACCEPTANCE CRITERIA      | <ul style="list-style-type: none"> <li>✓ Index stage runs only after successful embedding stage.</li> <li>✓ Vector dimension matches the configured embedding model.</li> <li>✓ Index writes are idempotent for the same chunk/version/model.</li> <li>✓ Index failure persists a typed stage error and prevents a false completed status.</li> </ul> |      |    |
| ARCHITECTURAL LAYER      | Database + Backend Core                                                                                                                                                                                                                                                                                                                               |      |    |
| ING-006                  | Idempotent re-ingestion & versioning                                                                                                                                                                                                                                                                                                                  | 2 SP | P0 |
| USER STORY               | As an admin, I want re-ingestion to be idempotent so that retries do not duplicate corpus state.                                                                                                                                                                                                                                                      |      |    |
| TECHNICAL / API CONTRACT | Use content hash + source/version keys; model active/inactive document versions explicitly.                                                                                                                                                                                                                                                           |      |    |
| ACCEPTANCE CRITERIA      | <ul style="list-style-type: none"> <li>✓ Same content/version creates no duplicate active chunks.</li> <li>✓ Changed content creates a new version lineage.</li> <li>✓ Old version remains queryable only when explicitly requested.</li> <li>✓ Duplicate concurrent jobs converge on one final corpus state.</li> </ul>                              |      |    |
| ARCHITECTURAL LAYER      | Backend Core + Database                                                                                                                                                                                                                                                                                                                               |      |    |

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|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----|
| ING-007                  | Ingestion dashboard & failure reporting                                                                                                                                                                                                                                                                      | 3 SP | P0 |
| USER STORY               | As an enterprise expert, I want to see ingestion status so that I can tell whether the corpus is trustworthy.                                                                                                                                                                                                |      |    |
| TECHNICAL / API CONTRACT | Dashboard with stage progress, per-document status, failure detail, retry control and corpus KPIs.                                                                                                                                                                                                           |      |    |
| ACCEPTANCE CRITERIA      | <ul style="list-style-type: none"> <li>✓ Shows queued/running/completed/failed states.</li> <li>✓ Failure row exposes stage + reason + retry action.</li> <li>✓ UI never reports success before indexing is confirmed.</li> <li>✓ Stage progress is emitted as live events, not a frozen spinner.</li> </ul> |      |    |
| ARCHITECTURAL LAYER      | Frontend UI                                                                                                                                                                                                                                                                                                  |      |    |

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|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----|
| ING-008                  | Corpus seeder: >=30 docs / >=150 pages                                                                                                                                                                                                                                                                  | 2 SP | P0 |
| USER STORY               | As an evaluator, I want a reproducible public/synthetic corpus so that the required evidence base can be demonstrated consistently.                                                                                                                                                                     |      |    |
| TECHNICAL / API CONTRACT | Add versioned seed manifest, import command and corpus validation script; use only public or synthetic data.                                                                                                                                                                                            |      |    |
| ACCEPTANCE CRITERIA      | <ul style="list-style-type: none"> <li>✓ Seeded corpus contains at least 30 documents and 150 pages.</li> <li>✓ Corpus validation fails below the floor.</li> <li>✓ No real personal data is included.</li> <li>✓ Seed output records document/version/page counts for README demo evidence.</li> </ul> |      |    |
| ARCHITECTURAL LAYER      | DevOps + Database                                                                                                                                                                                                                                                                                       |      |    |

| EPIC 02                                                                                                      | Grounded Hybrid Retrieval Engine | 6 stories |
|--------------------------------------------------------------------------------------------------------------|----------------------------------|-----------|
| Dense + keyword retrieval, documented fusion, traceable citations, refusal and measurable retrieval quality. |                                  |           |

| RET-001                  | Hybrid dense + keyword retriever | 3 SP                                                                                                                                                                                                                                                                                                      | P0 |
|--------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| USER STORY               |                                  | As a user, I want hybrid search so that semantic meaning and exact terminology both contribute to retrieval.                                                                                                                                                                                              |    |
| TECHNICAL / API CONTRACT |                                  | Dense pgvector search + PostgreSQL full-text/keyword search; deterministic fusion method and top-k configuration.                                                                                                                                                                                         |    |
| ACCEPTANCE CRITERIA      |                                  | <ul style="list-style-type: none"> <li>✓ Both channels run for a normal query.</li> <li>✓ Fusion produces one ranked list with documented scoring.</li> <li>✓ Top-k and weights are configuration, not hidden constants.</li> <li>✓ Retrieval trace stores both channel results before fusion.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                                  | AI/Agent Engine + Database                                                                                                                                                                                                                                                                                |    |

| RET-002                  | Retrieval enhancement: metadata filtering | 2 SP                                                                                                                                                                                                                                                                                                  | P0 |
|--------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| USER STORY               |                                           | As an expert, I want retrieval to respect metadata scope so that answers use the correct document version and section.                                                                                                                                                                                |    |
| TECHNICAL / API CONTRACT |                                           | Support source, document, version, section/page and language filters; reject incompatible scope.                                                                                                                                                                                                      |    |
| ACCEPTANCE CRITERIA      |                                           | <ul style="list-style-type: none"> <li>✓ No chunk outside filter scope is returned.</li> <li>✓ Version filter prevents stale evidence from being selected.</li> <li>✓ Applied filters appear in the trace.</li> <li>✓ Tests prove filter leakage is impossible within the supported scope.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                                           | AI/Agent Engine                                                                                                                                                                                                                                                                                       |    |

| RET-003                  | Structured citations & source drawer | 3 SP                                                                                                                                                                                                                                                                                                                   | P0 |
|--------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| USER STORY               |                                      | As an expert, I want every factual answer claim to link to an exact chunk so that I can verify it.                                                                                                                                                                                                                     |    |
| TECHNICAL / API CONTRACT |                                      | Citation object contains citationId, chunkId, documentId, version, page/clause, excerpt and source label; UI opens source drawer.                                                                                                                                                                                      |    |
| ACCEPTANCE CRITERIA      |                                      | <ul style="list-style-type: none"> <li>✓ Every grounded factual sentence/bullet has at least one citation.</li> <li>✓ Citation resolves to the exact stored chunk.</li> <li>✓ Page/clause is shown when available.</li> <li>✓ Citations cannot be fabricated by the UI when missing in the response schema.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                                      | Frontend UI + AI/Agent Engine                                                                                                                                                                                                                                                                                          |    |

| RET-004                  | Low-evidence refusal mechanism | 3 SP                                                                                                                                                                                                                                                                                                                              | P0 |
|--------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| USER STORY               |                                | As a user, I want the copilot to refuse when the corpus lacks evidence so that it does not guess.                                                                                                                                                                                                                                 |    |
| TECHNICAL / API CONTRACT |                                | Apply evidence threshold and/or support score gate before answer generation; typed refusal response.                                                                                                                                                                                                                              |    |
| ACCEPTANCE CRITERIA      |                                | <ul style="list-style-type: none"> <li>✓ Out-of-corpus query returns explicit refusal.</li> <li>✓ Refusal contains no unsupported factual claims.</li> <li>✓ Ambiguous low-evidence query escalates or asks for scope rather than guessing.</li> <li>✓ Threshold and decision are recorded in evaluation/trace output.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                                | AI/Agent Engine                                                                                                                                                                                                                                                                                                                   |    |

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|--------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| RET-005                  | Retrieval trace & debug inspector | 2 SP                                                                                                                                                                                                                                                                                                                      | P0 |
| USER STORY               |                                   | As an engineer, I want to inspect retrieval so that failures can be diagnosed after the fact.                                                                                                                                                                                                                             |    |
| TECHNICAL / API CONTRACT |                                   | Persist query, filters, dense top-k, keyword top-k, fusion, selected chunks and citation mapping.                                                                                                                                                                                                                         |    |
| ACCEPTANCE CRITERIA      |                                   | <ul style="list-style-type: none"> <li>✓ Run view shows considered and selected chunks.</li> <li>✓ Trace payload excludes secrets and raw credentials.</li> <li>✓ A single runId/correlationId links retrieval to downstream agent work.</li> <li>✓ Inspector can explain why a returned source ranked highly.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                                   | Frontend UI + AI/Agent Engine                                                                                                                                                                                                                                                                                             |    |

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|--------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| RET-006                  | Retrieval integration & regression tests | 2 SP                                                                                                                                                                                                                                                                                              | P0 |
| USER STORY               |                                          | As a developer, I want automated retrieval tests so that regressions are caught before merge.                                                                                                                                                                                                     |    |
| TECHNICAL / API CONTRACT |                                          | Version-aware fixture corpus; test top-k hit, fusion, filtering, refusal and citation linkage.                                                                                                                                                                                                    |    |
| ACCEPTANCE CRITERIA      |                                          | <ul style="list-style-type: none"> <li>✓ CI fails when expected evidence leaves the required top-k.</li> <li>✓ Version-aware cases distinguish old/new documents.</li> <li>✓ Refusal case passes when evidence floor is not met.</li> <li>✓ Tests run without external LLM dependency.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                                          | Backend Core + DevOps                                                                                                                                                                                                                                                                             |    |

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| EPIC 03                                                                                                                                   | Multi-Agent Workflow & Orchestration | 6 stories |
| Supervisor + exactly the three specialists required by the assigned Domain Pack, typed contracts, restricted tools and safe side effects. |                                      |           |

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|--------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| AGT-001                  | Typed agent contract framework | 3 SP                                                                                                                                                                                                                                                                    | P0 |
| USER STORY               |                                | As an engineer, I want typed input/output contracts so that agents communicate predictably.                                                                                                                                                                             |    |
| TECHNICAL / API CONTRACT |                                | Use discriminated unions / JSON Schema or equivalent typed contracts for each agent request, result, tool call and termination signal.                                                                                                                                  |    |
| ACCEPTANCE CRITERIA      |                                | <ul style="list-style-type: none"> <li>✓ Invalid agent output is rejected before downstream execution.</li> <li>✓ Schemas are versioned.</li> <li>✓ Contract tests run in CI.</li> <li>✓ Agents exchange structured data, never undocumented free-form text.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                                | AI/Agent Engine                                                                                                                                                                                                                                                         |    |

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| AGT-002                  | Supervisor/orchestrator state machine | 3 SP                                                                                                                                                                                                                                                                           | P0 |
| USER STORY               |                                       | As a system, I want a named orchestration pattern so that workflow control is explicit and inspectable.                                                                                                                                                                        |    |
| TECHNICAL / API CONTRACT |                                       | Implement supervisor state machine; max iterations, per-step timeout, retry/backoff, cancellation and graceful plain-RAG degradation.                                                                                                                                          |    |
| ACCEPTANCE CRITERIA      |                                       | <ul style="list-style-type: none"> <li>✓ Run terminates on max-iteration breaker.</li> <li>✓ Each step has bounded timeout.</li> <li>✓ Retry policy is visible in trace.</li> <li>✓ On agent failure the system either degrades safely or stops with a typed error.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                                       | AI/Agent Engine                                                                                                                                                                                                                                                                |    |

| AGT-003                  | Domain-specialist Agent 1 | 3 SP                                                                                                                                                                                                                                                                                                                                             | P0 |
|--------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| USER STORY               |                           | As a domain expert, I want the first assigned specialist to perform its exact domain role so that the workflow starts from structured evidence.                                                                                                                                                                                                  |    |
| TECHNICAL / API CONTRACT |                           | Replace the generic "Agent A" with the role named by the assigned Domain Pack in the ITI brief; restrict tools and define termination.                                                                                                                                                                                                           |    |
| ACCEPTANCE CRITERIA      |                           | <ul style="list-style-type: none"> <li>✓ Role name, inputs, outputs and termination condition match the assigned domain workflow.</li> <li>✓ Agent can use only its allow-listed tools.</li> <li>✓ Agent returns structured evidence/results with source references.</li> <li>✓ Prompt is a versioned artifact, not an inline string.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                           | AI/Agent Engine                                                                                                                                                                                                                                                                                                                                  |    |

| AGT-004                  | Domain-specialist Agent 2 | 3 SP                                                                                                                                                                                                                                                                                                  | P0 |
|--------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| USER STORY               |                           | As a domain expert, I want the second assigned specialist to perform its exact domain analysis so that interpretation is isolated from retrieval.                                                                                                                                                     |    |
| TECHNICAL / API CONTRACT |                           | Implement the role specified by the assigned Domain Pack; input comes from typed upstream result and evidence references.                                                                                                                                                                             |    |
| ACCEPTANCE CRITERIA      |                           | <ul style="list-style-type: none"> <li>✓ Agent rejects missing/invalid upstream evidence.</li> <li>✓ Tool set is narrower than orchestrator-wide tools.</li> <li>✓ Output includes confidence/evidence linkage where appropriate.</li> <li>✓ Termination condition is explicit and tested.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                           | AI/Agent Engine                                                                                                                                                                                                                                                                                       |    |

| AGT-005                  | Domain-specialist Agent 3 / drafting agent | 3 SP                                                                                                                                                                                                                                                                                                                                | P0 |
|--------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| USER STORY               |                                            | As a reviewer, I want the final domain specialist to draft from structured evidence so that unsupported conclusions are not introduced.                                                                                                                                                                                             |    |
| TECHNICAL / API CONTRACT |                                            | Implement the third assigned specialist; drafting prompt consumes structured results and citation handles.                                                                                                                                                                                                                          |    |
| ACCEPTANCE CRITERIA      |                                            | <ul style="list-style-type: none"> <li>✓ Draft is generated only from approved structured evidence.</li> <li>✓ Missing evidence yields a refusal/escalation, not invented content.</li> <li>✓ Output schema is validated before approval request creation.</li> <li>✓ Agent has no direct write/side-effect tool access.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                                            | AI/Agent Engine                                                                                                                                                                                                                                                                                                                     |    |

| AGT-006                  | Tool registry + safe side-effect adapter | 3 SP                                                                                                                                                                                                                                                                                                                                 | P0 |
|--------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| USER STORY               |                                          | As a system, I want restricted tool calling so that consequential actions cannot bypass human approval.                                                                                                                                                                                                                              |    |
| TECHNICAL / API CONTRACT |                                          | Register $\geq 4$ tools, including $\geq 1$ side-effecting tool; enforce per-agent allow-lists and approval-gate precondition.                                                                                                                                                                                                       |    |
| ACCEPTANCE CRITERIA      |                                          | <ul style="list-style-type: none"> <li>✓ At least four tools exist in the registry.</li> <li>✓ Every tool call validates typed arguments.</li> <li>✓ Side-effecting tools are blocked unless an approved approval token exists.</li> <li>✓ Tool execution records tool name, actor/agent, args hash and outcome in trace.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                                          | Backend Core + AI/Agent Engine                                                                                                                                                                                                                                                                                                       |    |

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| <b>EPIC 04</b> | <b>Human-in-the-Loop Approval Gate</b> | <b>6 stories</b> |
|----------------|----------------------------------------|------------------|

Consequential work pauses for explicit approval, with approve/reject/edit-and-approve and a P0 audit trail.

|                                 |                                     |                                                                                                                                                                                                                                                                                                                                                            |           |
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| <b>HITL-001</b>                 | <b>Approval request persistence</b> | <b>2 SP</b>                                                                                                                                                                                                                                                                                                                                                | <b>P0</b> |
| <b>USER STORY</b>               |                                     | As an approver, I want consequential actions to enter a review queue so that I remain in control.                                                                                                                                                                                                                                                          |           |
| <b>TECHNICAL / API CONTRACT</b> |                                     | Persist approvalRequest with runId, proposed action, evidence refs, risk level, requester, reviewer, timestamps and current status.                                                                                                                                                                                                                        |           |
| <b>ACCEPTANCE CRITERIA</b>      |                                     | <ul style="list-style-type: none"> <li>✓ Any side-effecting action creates an approval record before execution.</li> <li>✓ Approval status is pending/approved/rejected/edit-approved and auditable.</li> <li>✓ Record references exact workflow step and proposed payload.</li> <li>✓ No approval record can be deleted through normal UI/API.</li> </ul> |           |
| <b>ARCHITECTURAL LAYER</b>      |                                     | Backend Core + Database                                                                                                                                                                                                                                                                                                                                    |           |

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| <b>HITL-002</b>                 | <b>Review queue UI</b> | <b>3 SP</b>                                                                                                                                                                                                                                                                                            | <b>P0</b> |
| <b>USER STORY</b>               |                        | As an approver, I want a prioritised queue so that I can act on pending work quickly.                                                                                                                                                                                                                  |           |
| <b>TECHNICAL / API CONTRACT</b> |                        | Queue table/card list with risk, requester, age, evidence count, proposed action and current status.                                                                                                                                                                                                   |           |
| <b>ACCEPTANCE CRITERIA</b>      |                        | <ul style="list-style-type: none"> <li>✓ Pending items are filterable and sortable.</li> <li>✓ High-risk items are visually distinct.</li> <li>✓ Reviewer sees only items they are authorised to act on.</li> <li>✓ Queue state refreshes live or on explicit refresh without losing edits.</li> </ul> |           |
| <b>ARCHITECTURAL LAYER</b>      |                        | Frontend UI                                                                                                                                                                                                                                                                                            |           |

|                                 |                     |                                                                                                                                                                                                                                                                                                             |           |
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| <b>HITL-003</b>                 | <b>Approve flow</b> | <b>2 SP</b>                                                                                                                                                                                                                                                                                                 | <b>P0</b> |
| <b>USER STORY</b>               |                     | As an approver, I want to approve a proposed action so that the system can continue with explicit authorisation.                                                                                                                                                                                            |           |
| <b>TECHNICAL / API CONTRACT</b> |                     | Approve endpoint validates reviewer permission, state version and approval token; resumes workflow exactly once.                                                                                                                                                                                            |           |
| <b>ACCEPTANCE CRITERIA</b>      |                     | <ul style="list-style-type: none"> <li>✓ Approved action proceeds exactly once.</li> <li>✓ Second approval attempt is rejected idempotently.</li> <li>✓ Approval stores actor, timestamp and reason/comment if supplied.</li> <li>✓ Side-effect tool receives approval context before execution.</li> </ul> |           |
| <b>ARCHITECTURAL LAYER</b>      |                     | Frontend UI + Backend Core                                                                                                                                                                                                                                                                                  |           |

|                                 |                              |                                                                                                                                                                                                                                                                                                                                        |           |
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| <b>HITL-004</b>                 | <b>Edit-and-approve flow</b> | <b>3 SP</b>                                                                                                                                                                                                                                                                                                                            | <b>P0</b> |
| <b>USER STORY</b>               |                              | As an approver, I want to edit the proposed output and then approve it so that the final action reflects my judgement.                                                                                                                                                                                                                 |           |
| <b>TECHNICAL / API CONTRACT</b> |                              | Editable structured fields plus comment; revalidate schema and risk guardrails before approval.                                                                                                                                                                                                                                        |           |
| <b>ACCEPTANCE CRITERIA</b>      |                              | <ul style="list-style-type: none"> <li>✓ Edited content is stored as a separate approved version.</li> <li>✓ Schema and domain guardrails run again after editing.</li> <li>✓ Audit shows original vs approved version hashes or snapshots.</li> <li>✓ Workflow resumes using the approved version, not the original draft.</li> </ul> |           |
| <b>ARCHITECTURAL LAYER</b>      |                              | Frontend UI + Backend Core                                                                                                                                                                                                                                                                                                             |           |

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| <b>HITL-005</b> | <b>Reject flow with reason</b> | <b>2 SP</b> | <b>P0</b> |
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| USER STORY               | As an approver, I want to reject a proposal with a reason so that unsafe work stops and the failure is learnable.                                                                                                                                                              |
| TECHNICAL / API CONTRACT | Reject endpoint with mandatory reason, terminal or rework state and trace linkage.                                                                                                                                                                                             |
| ACCEPTANCE CRITERIA      | <ul style="list-style-type: none"> <li>✓ Reject requires a non-empty reason.</li> <li>✓ Rejected side effect is never executed.</li> <li>✓ Workflow records whether it terminated or entered a rework path.</li> <li>✓ Reason is visible in the run audit timeline.</li> </ul> |
| ARCHITECTURAL LAYER      | Frontend UI + Backend Core                                                                                                                                                                                                                                                     |

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| HITL-006                 | Approval audit timeline                                                                                                                                                                                                                                                                                                                                      | 2 SP | P0 |
| USER STORY               | As an auditor, I want to see the full decision history so that human control is provable.                                                                                                                                                                                                                                                                    |      |    |
| TECHNICAL / API CONTRACT | Immutable-style append-only approval events and timeline UI with actor, action, timestamp, previous/current state and payload hash.                                                                                                                                                                                                                          |      |    |
| ACCEPTANCE CRITERIA      | <ul style="list-style-type: none"> <li>✓ All approve/reject/edit-and-approve actions appear in order.</li> <li>✓ Events include actor identity and run/correlation IDs.</li> <li>✓ Audit rows cannot be modified through normal application paths.</li> <li>✓ This story is P0 because the ITI brief requires all approval actions to be audited.</li> </ul> |      |    |
| ARCHITECTURAL LAYER      | Frontend UI + Database                                                                                                                                                                                                                                                                                                                                       |      |    |

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| EPIC 05                                                                                         | Real-time UI & Streaming | 4 stories |
| Token streaming, live workflow events, server-side cancellation and persistent session history. |                          |           |

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| RT-001                   | SSE streaming endpoint                                                                                                                                                                                                                                                              | 3 SP | P0 |
| USER STORY               | As a user, I want token-level streaming so that long answers feel live and transparent.                                                                                                                                                                                             |      |    |
| TECHNICAL / API CONTRACT | GET/POST streaming route with server-sent events for token, citation, status and error events; heartbeat and completion event.                                                                                                                                                      |      |    |
| ACCEPTANCE CRITERIA      | <ul style="list-style-type: none"> <li>✓ Tokens arrive incrementally.</li> <li>✓ Completion event is emitted exactly once.</li> <li>✓ Connection errors produce a typed terminal event where possible.</li> <li>✓ Server stops generation when the request is cancelled.</li> </ul> |      |    |
| ARCHITECTURAL LAYER      | Backend Core                                                                                                                                                                                                                                                                        |      |    |

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| RT-002                   | Live agent progress events                                                                                                                                                                                                                                          | 3 SP | P0 |
| USER STORY               | As a user, I want to see which agent is running so that the workflow does not look frozen.                                                                                                                                                                          |      |    |
| TECHNICAL / API CONTRACT | Emit agent_started, tool_started, retrieval_started, approval_pending, agent_completed and failure events with timestamps.                                                                                                                                          |      |    |
| ACCEPTANCE CRITERIA      | <ul style="list-style-type: none"> <li>✓ UI shows current step and completed steps.</li> <li>✓ Events are linked to runId/correlationId.</li> <li>✓ No fake progress timers are used.</li> <li>✓ Trace and live stream share the same event identifiers.</li> </ul> |      |    |
| ARCHITECTURAL LAYER      | Backend Core + Frontend UI                                                                                                                                                                                                                                          |      |    |

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| RT-003     | Client cancellation                                                                    | 3 SP | P0 |
| USER STORY | As a user, I want to cancel a run so that I can stop unnecessary work and token spend. |      |    |

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| TECHNICAL / API CONTRACT | AbortController/client cancel propagated to server cancellation token; orchestrator halts future work and provider stream.                                                                                                                                                                                               |
| ACCEPTANCE CRITERIA      | <ul style="list-style-type: none"> <li>✓ Cancel stops server-side workflow, not only browser rendering.</li> <li>✓ In-flight provider request is cancelled where supported.</li> <li>✓ Run is marked cancelled and no later agent executes.</li> <li>✓ Cancellation is visible in the trace and token ledger.</li> </ul> |
| ARCHITECTURAL LAYER      | Frontend UI + Backend Core                                                                                                                                                                                                                                                                                               |

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| RT-004                   | Persistent session history                                                                                                                                                                                                                                               | 2 SP | P0 |
| USER STORY               | As a user, I want past conversations and runs to remain available so that work can be resumed.                                                                                                                                                                           |      |    |
| TECHNICAL / API CONTRACT | Persist sessions, messages, run references and last-known status; support reload and revisit.                                                                                                                                                                            |      |    |
| ACCEPTANCE CRITERIA      | <ul style="list-style-type: none"> <li>✓ History survives page reload and server restart.</li> <li>✓ Users can only access authorised sessions.</li> <li>✓ Large traces are loaded on demand.</li> <li>✓ Session entries link to their associated run traces.</li> </ul> |      |    |
| ARCHITECTURAL LAYER      | Database + Frontend UI                                                                                                                                                                                                                                                   |      |    |

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| EPIC 06                                                                                                                                     | Mandatory Twist Implementation | 6 stories |
| Lock the assigned D<n>/T<n>, implement the exact domain workflow/risk, and keep the twist measurable and isolated from core business logic. |                                |           |

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| TW-001                   | Variant lock & domain configuration                                                                                                                                                                                                                                                           | 2 SP | P0 |
| USER STORY               | As a solution architect, I want the assigned Domain and Twist encoded as explicit configuration so that the MVP cannot silently drift from the invitation.                                                                                                                                    |      |    |
| TECHNICAL / API CONTRACT | Create immutable runtime configuration: ASSIGNED_DOMAIN and ASSIGNED_TWIST; fail fast if placeholders remain in production/demo mode.                                                                                                                                                         |      |    |
| ACCEPTANCE CRITERIA      | <ul style="list-style-type: none"> <li>✓ Application clearly reports the active D?/T? values.</li> <li>✓ Placeholder values block release/demo mode.</li> <li>✓ README states the derivation or invitation source.</li> <li>✓ No code path silently falls back to another variant.</li> </ul> |      |    |
| ARCHITECTURAL LAYER      | Backend Core + DevOps                                                                                                                                                                                                                                                                         |      |    |

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| TW-002                   | Domain workflow & risk guard                                                                                                                                                                                                                                                                                          | 3 SP | P0 |
| USER STORY               | As an evaluator, I want the assigned Domain Pack workflow and its named risk to be encoded as executable guardrails so that the platform addresses the variant-specific failure mode.                                                                                                                                 |      |    |
| TECHNICAL / API CONTRACT | Encode the exact workflow sequence and the Domain Pack risk guard from the ITI brief; make high-risk constraints structural where the brief demands it.                                                                                                                                                               |      |    |
| ACCEPTANCE CRITERIA      | <ul style="list-style-type: none"> <li>✓ Three specialist roles match the assigned Domain Pack.</li> <li>✓ Workflow stages match the stated arrow sequence.</li> <li>✓ Named risk guard has a deterministic enforcement point.</li> <li>✓ A negative test proves the guard blocks the prohibited behavior.</li> </ul> |      |    |
| ARCHITECTURAL LAYER      | AI/Agent Engine + Backend Core                                                                                                                                                                                                                                                                                        |      |    |

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| TW-003 | Twist acceptance contract | 3 SP | P0 |
|--------|---------------------------|------|----|

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| USER STORY               | As an evaluator, I want the mandatory twist to have explicit measurable tests so that compliance is demonstrable.                                                                                                                                       |
| TECHNICAL / API CONTRACT | Define twist contract, input/output invariants, dedicated tests and a demo scenario; keep capability boundaries honest.                                                                                                                                 |
| ACCEPTANCE CRITERIA      | <ul style="list-style-type: none"> <li>✓ Twist core behavior has at least one end-to-end test.</li> <li>✓ Failure path is tested.</li> <li>✓ Twist metric/report is produced by the harness.</li> <li>✓ README demo path includes the twist.</li> </ul> |
| ARCHITECTURAL LAYER      | Backend Core + AI/Agent Engine                                                                                                                                                                                                                          |

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| TW-004                   | Twist-specific implementation adapter | 3 SP                                                                                                                                                                                                                                                             | P0 |
| USER STORY               |                                       | As an engineer, I want the assigned Twist implemented behind a narrow adapter so that its complexity does not leak into domain logic.                                                                                                                            |    |
| TECHNICAL / API CONTRACT |                                       | Create one adapter/service boundary for the assigned T<n>; select it by configuration.                                                                                                                                                                           |    |
| ACCEPTANCE CRITERIA      |                                       | ✓ Domain/application layers depend on a twist port, not provider/framework SDKs.<br>✓ Twist adapter can be stubbed in unit tests.<br>✓ Replacing the twist implementation does not alter business rules.<br>✓ The central twist decision is captured in ADR-004. |    |
| ARCHITECTURAL LAYER      |                                       | AI/Agent Engine + Backend Core                                                                                                                                                                                                                                   |    |

| TW-005                   | Twist-specific UI surface | 2 SP                                                                                                                                                                                                                                                                                      | P0 |
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| USER STORY               |                           | As a user, I want the mandatory Twist visible in the workflow so that its behavior is observable rather than hidden.                                                                                                                                                                      |    |
| TECHNICAL / API CONTRACT |                           | Add only the UI controls/status needed by the assigned Twist; keep the reference visual language.                                                                                                                                                                                         |    |
| ACCEPTANCE CRITERIA      |                           | <div>✓ UI exposes the minimum control/status needed to exercise the Twist.</div> <div>✓ Twist state is represented in the trace or task context.</div> <div>✓ No UI-only mock is accepted as implementation evidence.</div> <div>✓ Responsive fallback exists for the core control.</div> |    |
| ARCHITECTURAL LAYER      |                           | Frontend UI                                                                                                                                                                                                                                                                               |    |

| TW-006                   | Twist observability & evaluation slice | 2 SP                                                                                                                                                                                                                           | P0 |
|--------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| USER STORY               |                                        | As an evaluator, I want twist-specific metrics visible so that compliance is proven with measurements.                                                                                                                         |    |
| TECHNICAL / API CONTRACT |                                        | Add a benchmark subset, result columns, threshold and dashboard/CLI output for the twist.                                                                                                                                      |    |
| ACCEPTANCE CRITERIA      |                                        | ✓ Harness runs twist subset independently.<br>✓ Actual pass/fail numbers are captured.<br>✓ Bad cases are retained and interpreted in docs/EVALUATION.md.<br>✓ Product demo can show one passing and one failing/guarded case. |    |
| ARCHITECTURAL LAYER      |                                        | Observability + Evaluation                                                                                                                                                                                                     |    |

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| EPIC 07                                                                                                              | Observability, Cost Tracking & LLM Evaluation Harness | 7 stories |
| End-to-end traceability, token/cost accounting, 25+ Q/A evaluation, adversarial testing and honest failure analysis. |                                                       |           |

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| OBS-001                  | Correlation ID propagation | 2 SP                                                                                                                                                                                                                                                                                                                                                                        | P0 |
| USER STORY               |                            | As an engineer, I want one correlation ID from HTTP request to agent/tool/LLM spans so that incidents are traceable end-to-end.                                                                                                                                                                                                                                             |    |
| TECHNICAL / API CONTRACT |                            | Create or accept correlation ID at API boundary; persist trace spans/events in the run trace store and propagate through orchestrator, agents, tools and providers.                                                                                                                                                                                                         |    |
| ACCEPTANCE CRITERIA      |                            | <ul style="list-style-type: none"> <li>✓ Every span/run has correlationId.</li> <li>✓ Missing inbound ID is generated.</li> <li>✓ UI trace exposes the identifier.</li> <li>✓ Persisted spans/events can be joined by correlationId after the run completes.</li> <li>✓ Trace can identify the request -&gt; orchestrator -&gt; agent -&gt; tool -&gt; LLM path.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                            | Backend Core + Observability                                                                                                                                                                                                                                                                                                                                                |    |

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| OBS-002                  | Token & cost ledger | 3 SP                                                                                                                                                                                                                                                                                                 | P0 |
| USER STORY               |                     | As an admin, I want per-request token and cost accounting so that usage is measurable and controllable.                                                                                                                                                                                              |    |
| TECHNICAL / API CONTRACT |                     | Persist provider/model/input/output/total tokens, unit pricing, estimated and actual cost, runId and correlationId.                                                                                                                                                                                  |    |
| ACCEPTANCE CRITERIA      |                     | <ul style="list-style-type: none"> <li>✓ Each provider call creates a usage row.</li> <li>✓ Run totals reconcile with call rows within documented rounding.</li> <li>✓ Unknown pricing is flagged rather than silently zeroed.</li> <li>✓ Spend can be queried by user/date/run/provider.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                     | Database + Observability                                                                                                                                                                                                                                                                             |    |

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| OBS-003                  | Trace inspector UI | 3 SP                                                                                                                                                                                                                                                                                          | P0 |
| USER STORY               |                    | As an evaluator, I want to inspect a completed run so that I can see which agents, tools, chunks and costs were involved.                                                                                                                                                                     |    |
| TECHNICAL / API CONTRACT |                    | Nested timeline for retrieval, agents, tools, approvals and LLM calls; no secrets.                                                                                                                                                                                                            |    |
| ACCEPTANCE CRITERIA      |                    | <ul style="list-style-type: none"> <li>✓ A run is understandable without server logs.</li> <li>✓ Clicking retrieval reveals chunk/citation links.</li> <li>✓ Clicking an LLM span reveals model/tokens/cost but not secrets.</li> <li>✓ Trace can be opened by runId from history.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                    | Frontend UI + Observability                                                                                                                                                                                                                                                                   |    |

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| OBS-004                  | Golden Q/A evaluation harness | 3 SP                                                                                                                                                                                                                                                                                                                                                                         | P0 |
| USER STORY               |                               | As an engineer, I want a runnable benchmark so that retrieval and groundedness can be measured before submission.                                                                                                                                                                                                                                                            |    |
| TECHNICAL / API CONTRACT |                               | Seed >=25 Q/A pairs with >=5 adversarial cases; report retrieval hit-rate, groundedness and refusal correctness.                                                                                                                                                                                                                                                             |    |
| ACCEPTANCE CRITERIA      |                               | <ul style="list-style-type: none"> <li>✓ At least 25 golden pairs exist.</li> <li>✓ At least 5 adversarial cases include out-of-corpus, ambiguous, prompt-injection and conflicting-source patterns.</li> <li>✓ Harness exits non-zero on critical schema/metric failures.</li> <li>✓ Actual baseline numbers, including bad ones, are saved with interpretation.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                               | Evaluation + DevOps                                                                                                                                                                                                                                                                                                                                                          |    |

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| OBS-005    | Prompt injection regression suite | 2 SP                                                             | P0 |
| USER STORY |                                   | As a security-minded engineer, I want direct and indirect prompt |    |

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|                          | injection tests so that retrieved text cannot silently rewrite agent instructions.                                                                                                                                                                                                                                         |
| TECHNICAL / API CONTRACT | Include >=3 injection cases; one indirect injection lives inside a retrieved document; verify privilege separation and tool restrictions.                                                                                                                                                                                  |
| ACCEPTANCE CRITERIA      | <ul style="list-style-type: none"> <li>✓ Injected document text is treated as untrusted content.</li> <li>✓ No injected content changes system/agent policy.</li> <li>✓ Side-effecting tool cannot run from injected instructions.</li> <li>✓ Results are included in docs/EVALUATION.md and security evidence.</li> </ul> |
| ARCHITECTURAL LAYER      | Security + Evaluation                                                                                                                                                                                                                                                                                                      |

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| OBS-006                  | Health/readiness endpoints                                                                                                                                                                                            | 1 SP | P0 |
| USER STORY               | As an operator, I want health and readiness endpoints so that Docker and CI can detect system health.                                                                                                                 |      |    |
| TECHNICAL / API CONTRACT | Implement /healthz and /readyz; readiness checks DB/vector connectivity and critical dependencies.                                                                                                                    |      |    |
| ACCEPTANCE CRITERIA      | <ul style="list-style-type: none"> <li>✓ Health returns process liveness.</li> <li>✓ Readiness fails when required dependency is unavailable.</li> <li>✓ Compose startup can use readiness before seeding.</li> </ul> |      |    |
| ARCHITECTURAL LAYER      | Backend Core + DevOps                                                                                                                                                                                                 |      |    |

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| OBS-007                  | Evaluation report & failure analysis                                                                                                                                                                                                                                    | 2 SP | P0 |
| USER STORY               | As an evaluator, I want a reproducible report of what worked and failed so that the submission is evidence-based.                                                                                                                                                       |      |    |
| TECHNICAL / API CONTRACT | Generate docs/EVALUATION.md from a repeatable command with metrics, baselines, adversarial results, limitations and follow-up actions.                                                                                                                                  |      |    |
| ACCEPTANCE CRITERIA      | <ul style="list-style-type: none"> <li>✓ Report includes actual numbers, not target numbers.</li> <li>✓ Every failed test has an interpretation or mitigation.</li> <li>✓ Report identifies corpus/eval version.</li> <li>✓ Command is documented in README.</li> </ul> |      |    |
| ARCHITECTURAL LAYER      | Documentation + Evaluation                                                                                                                                                                                                                                              |      |    |

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| EPIC 08                                                                                                                                              | Clean Architecture, Dockerization & DevOps | 15 stories |
| Architecture boundaries, providers, auth/RBAC, OpenAPI, tests, Docker, CI/security, mandatory documentation, AI workflow and Git/process discipline. |                                            |            |

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| DEV-001                  | Clean architecture module boundaries                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3 SP | P0 |
| USER STORY               | As an engineer, I want domain/application code independent of infrastructure so that providers can be swapped safely.                                                                                                                                                                                                                                                                                                                                                                              |      |    |
| TECHNICAL / API CONTRACT | Use Clean/Hexagonal/Onion/Vertical Slice; define Domain -> Application -> Adapters/Infrastructure -> API boundaries; use dependency injection throughout and explicit domain/application error types.                                                                                                                                                                                                                                                                                              |      |    |
| ACCEPTANCE CRITERIA      | <ul style="list-style-type: none"> <li>✓ Domain and application layers import no LLM/vector/web framework SDKs.</li> <li>✓ Architecture test detects forbidden imports.</li> <li>✓ One adapter replacement demonstrates business logic remains unchanged.</li> <li>✓ Application services and adapters are resolved through DI, not manually constructed in domain/application code.</li> <li>✓ Domain/application failures use explicit typed error codes mapped to stable API errors.</li> </ul> |      |    |

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| ARCHITECTURAL LAYER      |                                                  | Backend Core                                                                                                                                                                                                                                                                                                          |    |
| DEV-002                  | Unified AI provider abstraction with 2 providers | 3 SP                                                                                                                                                                                                                                                                                                                  | P0 |
| USER STORY               |                                                  | As an architect, I want a provider interface covering completion, streaming, tool calling and embeddings so that the platform survives provider changes.                                                                                                                                                              |    |
| TECHNICAL / API CONTRACT |                                                  | Define completion, streaming, tool-call and embedding ports; implement >=2 working providers via configuration with fallback chain.                                                                                                                                                                                   |    |
| ACCEPTANCE CRITERIA      |                                                  | <ul style="list-style-type: none"> <li>✓ Hosted provider path works.</li> <li>✓ Alternative/local provider path works.</li> <li>✓ Provider chosen by configuration, not business logic.</li> <li>✓ Fallback behavior and capability differences are documented honestly.</li> </ul>                                   |    |
| ARCHITECTURAL LAYER      |                                                  | AI/Agent Engine                                                                                                                                                                                                                                                                                                       |    |
| DEV-003                  | Server-side authentication & RBAC                | 3 SP                                                                                                                                                                                                                                                                                                                  | P0 |
| USER STORY               |                                                  | As a user, I want secure authentication and role-based permissions so that only authorised actors can access or approve work.                                                                                                                                                                                         |    |
| TECHNICAL / API CONTRACT |                                                  | Implement login/session or JWT; define >=2 genuinely different roles; enforce authorization on server routes and object ownership.                                                                                                                                                                                    |    |
| ACCEPTANCE CRITERIA      |                                                  | <ul style="list-style-type: none"> <li>✓ Unauthenticated protected requests fail.</li> <li>✓ Two roles have materially different permissions.</li> <li>✓ Object ownership is checked for documents/sessions/runs/approvals.</li> <li>✓ Negative tests prove UI button hiding is not the security boundary.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                                                  | Backend Core + Database                                                                                                                                                                                                                                                                                               |    |
| DEV-004                  | OpenAPI contract & HTTP documentation            | 2 SP                                                                                                                                                                                                                                                                                                                  | P0 |
| USER STORY               |                                                  | As an integrator, I want a documented HTTP API so that the platform can be understood and tested independently of the UI.                                                                                                                                                                                             |    |
| TECHNICAL / API CONTRACT |                                                  | Generate/maintain OpenAPI specification with auth schemes, request/response schemas, errors, streaming endpoints and examples.                                                                                                                                                                                        |    |
| ACCEPTANCE CRITERIA      |                                                  | <ul style="list-style-type: none"> <li>✓ All MVP endpoints appear in OpenAPI.</li> <li>✓ Schemas match runtime validation.</li> <li>✓ Authentication and role requirements are documented.</li> <li>✓ README links to API docs.</li> </ul>                                                                            |    |
| ARCHITECTURAL LAYER      |                                                  | Backend Core + Documentation                                                                                                                                                                                                                                                                                          |    |
| DEV-005                  | Test pyramid & contract tests                    | 3 SP                                                                                                                                                                                                                                                                                                                  | P0 |
| USER STORY               |                                                  | As a maintainer, I want sharp tests around domain, retrieval and agent contracts so that regressions are cheap to find.                                                                                                                                                                                               |    |
| TECHNICAL / API CONTRACT |                                                  | Unit tests with LLM stubs; integration tests for ingestion/retrieval; contract tests for tool/agent schemas; security negative tests.                                                                                                                                                                                 |    |
| ACCEPTANCE CRITERIA      |                                                  | <ul style="list-style-type: none"> <li>✓ Tests run offline where possible.</li> <li>✓ At least 10 non-trivial high-risk tests exist.</li> <li>✓ Agent/tool contract drift fails CI.</li> <li>✓ Tests cover approval bypass, cross-object access and refusal behavior.</li> </ul>                                      |    |
| ARCHITECTURAL LAYER      |                                                  | DevOps                                                                                                                                                                                                                                                                                                                |    |

| DEV-006                  | Docker Compose & seed command | 2 SP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | P0 |
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| USER STORY               |                               | As an evaluator, I want one reproducible command to run the system so that the repository is easy to assess on a clean machine.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |
| TECHNICAL / API CONTRACT |                               | docker compose up starts web, API and PostgreSQL/pgvector; provide committed migrations plus seed and ingest commands; keep configuration externalised in .env.example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |
| ACCEPTANCE CRITERIA      |                               | <ul style="list-style-type: none"> <li>✓ Fresh clone boots on a clean machine.</li> <li>✓ No manual hidden steps are required.</li> <li>✓ Seed creates demo accounts and corpus fixtures.</li> <li>✓ Readiness gates prevent ingestion before dependencies are ready.</li> <li>✓ Fresh PostgreSQL/pgvector schema is reproducible entirely from committed migrations.</li> <li>✓ Seed/ingest runs only after migrations and readiness checks succeed.</li> <li>✓ .env.example documents every required environment variable and contains no real secrets.</li> <li>✓ README treats Docker as the only prerequisite unless explicitly documented.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                               | DevOps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |

| DEV-007                  | CI, branch protection & secret scanning | 3 SP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | P0 |
|--------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| USER STORY               |                                         | As a maintainer, I want CI and repository controls to enforce engineering discipline so that main is always defensible.                                                                                                                                                                                                                                                                                                                                                                                                     |    |
| TECHNICAL / API CONTRACT |                                         | GitHub Actions on every PR: build, lint, tests, dependency scan, secret scan; branch protection; protected main; repository hygiene and release tags.                                                                                                                                                                                                                                                                                                                                                                       |    |
| ACCEPTANCE CRITERIA      |                                         | <ul style="list-style-type: none"> <li>✓ CI runs on every PR.</li> <li>✓ Dependency and secret scanning are enforced.</li> <li>✓ Main is protected from direct pushes.</li> <li>✓ Full Git history is scanned for secrets before submission.</li> <li>✓ Correct .gitignore is present from the initial repository commit.</li> <li>✓ README, LICENSE, CONTRIBUTING.md, .env.example, PR/issue templates and CODEOWNERS are present.</li> <li>✓ Release tags mark meaningful increments, including the final MVP.</li> </ul> |    |
| ARCHITECTURAL LAYER      |                                         | DevOps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |

| DEV-008                  | OWASP Web + LLM security controls | 3 SP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | P0 |
|--------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| USER STORY               |                                   | As a security reviewer, I want documented controls against web and LLM threats so that the system is defensible.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
| TECHNICAL / API CONTRACT |                                   | Implement object-ownership checks, parameterised queries, validated uploads, cryptographic protections for credentials, rate/abuse controls, request and payload limits, secure headers/CORS, tool arg validation, PII handling, prompt injection defense, token/iteration limits and pinned dependencies with a committed lockfile. Document what data leaves the infrastructure when an external provider is used.                                                                                                                                              |    |
| ACCEPTANCE CRITERIA      |                                   | <ul style="list-style-type: none"> <li>✓ &gt;=3 injection cases are demonstrably resisted.</li> <li>✓ No raw model output is rendered as executable HTML or sent unchecked to shell/SQL/file paths.</li> <li>✓ Sensitive logs never contain secrets.</li> <li>✓ Request/upload payload limits are enforced.</li> <li>✓ Credential handling uses approved password hashing / secret handling and is not stored plaintext.</li> <li>✓ PII is detected/redacted or prevented by the supported data boundary, and data egress to external LLM providers is</li> </ul> |    |

|                     |                                                                                                                                                                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | documented.<br>✓ Security headers and CORS are restrictive rather than permissive defaults.<br>✓ Pinned dependency versions and the committed lockfile are scanned in CI.<br>✓ Security controls are mapped to threats in docs/SECURITY.md. |
| ARCHITECTURAL LAYER | Security                                                                                                                                                                                                                                    |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |    |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----|
| DEV-009                  | BRD + System Design + Architecture docs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3 SP | P0 |
| USER STORY               | As an evaluator/instructor, I want the architecture rationale documented so that the system can be defended and taught.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |    |
| TECHNICAL / API CONTRACT | Create BRD, SYSTEM-DESIGN, ARCHITECTURE, and >=4 ADRs; include target vs implemented gap table and decision alternatives. Target architecture must explicitly consider gateway, managed rate limiting, secrets manager, broker/queue, autoscaling, caching, managed vector DB, observability stack, CI/CD environments, DR/backup and cost at scale.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |      |    |
| ACCEPTANCE CRITERIA      | ✓ BRD has personas, measurable objectives, unique requirements, assumptions, risks, out-of-scope and traceability matrix.<br>✓ System Design has Part A Target Architecture and Part B Implemented MVP with a gap table covering deferred items, mitigation, effort and cost to close.<br>✓ Target architecture addresses gateway/rate limiting, secrets, broker, autoscaling, caching, vector DB, observability, CI/CD, DR/backup and cost-at-scale.<br>✓ Architecture docs include C4 L1-L3, full workflow sequence including approval and streaming, data-flow trust boundaries, ER diagram and layer dependency diagram.<br>✓ Diagram source files are committed, not only rendered images.<br>✓ Four ADRs cover chunking/retrieval, orchestration, vector store and twist, with alternatives considered and rejected. |      |    |
| ARCHITECTURAL LAYER      | Documentation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |    |

|                          |                                                                                                                                                                                                                                                                             |      |    |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----|
| DEV-010                  | Agentic coding workflow pack                                                                                                                                                                                                                                                | 2 SP | P0 |
| USER STORY               | As an instructor, I want the AI-assisted development process governed and explainable so that AI is treated as a controlled engineering participant.                                                                                                                        |      |    |
| TECHNICAL / API CONTRACT | Commit docs/AGENTIC-WORKFLOW.md plus >=5 of: project instruction file, versioned prompt assets/skills, sub-agents, quality hooks, custom commands, MCP usage, versioned prompt library.                                                                                     |      |    |
| ACCEPTANCE CRITERIA      | ✓ At least five required workflow mechanisms are present.<br>✓ The document explains what was configured, why, impact and where the agentic approach failed.<br>✓ Architecture rules are encoded where practical.<br>✓ Prompts used by the product are versioned artifacts. |      |    |
| ARCHITECTURAL LAYER      | DevOps + Documentation                                                                                                                                                                                                                                                      |      |    |

|                          |                                                                                                                                                                  |      |    |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----|
| DEV-011                  | AI usage log & verification record                                                                                                                               | 1 SP | P0 |
| USER STORY               | As an evaluator, I want an honest AI usage log so that delegated work and model mistakes are auditable.                                                          |      |    |
| TECHNICAL / API CONTRACT | Maintain docs/AI-USAGE-LOG.md with delegated task, what was written/changed by the author, verification method, model mistakes/misleading output and correction. |      |    |
| ACCEPTANCE CRITERIA      | ✓ Log is updated throughout the project.<br>✓ At least one verification example documents a model error or correction when one occurs.                           |      |    |

|                          |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |
|--------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
|                          |                                              | ✓ Claims of flawless AI use are avoided unless supported.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
| ARCHITECTURAL LAYER      |                                              | Documentation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |
| DEV-012                  | Git story: >=30 commits / >=6 days / >=8 PRs | 2 SP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | P0 |
| USER STORY               |                                              | As a maintainer, I want the repository history to show deliberate engineering progress so that the submission is credible as an instructor artifact.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
| TECHNICAL / API CONTRACT |                                              | Plan atomic Conventional Commits (or documented equivalent); >=8 real PRs with what/why/how-tested descriptions and inline self-review comments; linked GitHub Issues; a board or milestones; protected main; release tags; disclose any starter/template repository in README and the first commit.                                                                                                                                                                                                                                                                                                                   |    |
| ACCEPTANCE CRITERIA      |                                              | ✓ At least 30 meaningful commits span at least 6 distinct days.<br>✓ At least 8 PRs have what/why/how-tested descriptions and self-review comments.<br>✓ All work merges through PRs; no direct pushes to main.<br>✓ Issues are linked from PRs where appropriate, and board/milestones show planned, active and deliberately deferred work.<br>✓ Commit messages follow the chosen convention and changes remain atomic.<br>✓ No WIP/fix2/final-final dump pattern.<br>✓ Any starter/template repository is declared in README and the first commit.                                                                  |    |
| ARCHITECTURAL LAYER      |                                              | DevOps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |
| DEV-013                  | Teaching pack & submission artifacts         | 3 SP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | P0 |
| USER STORY               |                                              | As an instructor, I want the teaching pack and submission artifacts ready from day one so that the repository can become a teachable lab.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
| TECHNICAL / API CONTRACT |                                              | Create teaching/ with 15-25 slides for a 90-minute postgraduate session, hands-on lab sheet, expected outputs, >=3 stretch challenges, answer key, learning-outcomes/assessment map, and a one-page common-trainee-mistakes note covering five predicted misconceptions and corrections; include README demo path and submission checklist.                                                                                                                                                                                                                                                                            |    |
| ACCEPTANCE CRITERIA      |                                              | ✓ Teaching pack covers one topic directly derived from the implementation.<br>✓ Session is designed for 90 minutes.<br>✓ Lab is runnable against the repository and includes expected outputs and answer key.<br>✓ At least three stretch challenges are included.<br>✓ Common-trainee-mistakes note covers five misconceptions with corrections.<br>✓ README contains 5-Minute Demo Path, prerequisites, every environment variable, free/local model path, seeded demo accounts, tests/evaluation commands and troubleshooting.<br>✓ Submission checklist covers both required videos and final repository evidence. |    |
| ARCHITECTURAL LAYER      |                                              | Documentation + Teaching                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |
| VIDEO-001                | Product demo video                           | 1 SP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | P0 |
| USER STORY               |                                              | As an evaluator, I want a concise product demo so that the core capabilities can be verified end-to-end.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |
| TECHNICAL / API CONTRACT |                                              | Record a 5-8 minute demo covering ingest, grounded answer with citations, correct refusal, multi-agent live progress, approval gate, trace and the assigned Twist.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |
| ACCEPTANCE CRITERIA      |                                              | ✓ Video is 5-8 minutes.<br>✓ It demonstrates every required capability named in the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |

|                          |                       |                                                                                                                                                                                                                                                            |    |
|--------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
|                          |                       | assessment.<br>✓ The assigned Twist is demonstrated, not merely described.<br>✓ Unlisted link is tested in a private browser session.<br>✓ README contains the validated link.                                                                             |    |
| ARCHITECTURAL LAYER      |                       | Documentation + QA                                                                                                                                                                                                                                         |    |
| VIDEO-002                | Teaching sample video | 1 SP                                                                                                                                                                                                                                                       | P0 |
| USER STORY               |                       | As an instructor, I want a teaching sample video so that my ability to teach the implemented topic is demonstrable.                                                                                                                                        |    |
| TECHNICAL / API CONTRACT |                       | Record a ~10-minute teaching segment based on the teaching pack; face and voice required; upload as an unlisted link and test it in a private browser session.                                                                                             |    |
| ACCEPTANCE CRITERIA      |                       | ✓ Video is about 10 minutes.<br>✓ Face and voice are present.<br>✓ It teaches a real slice of the implemented system rather than reading slides aloud.<br>✓ Unlisted link is tested in a private browser session.<br>✓ README contains the validated link. |    |
| ARCHITECTURAL LAYER      |                       | Teaching + Documentation                                                                                                                                                                                                                                   |    |

## PART 3 - UI Layout & Component Wireframe Specification

### 1. Ingestion Dashboard

|                |                                                                                                           |
|----------------|-----------------------------------------------------------------------------------------------------------|
| TOP BAR        | Navy banner; corpus selector; D?/T? badge; environment badge; Upload button.                              |
| KPI STRIP      | Documents, Pages, Indexed Chunks, Failures, Last Ingest; green healthy state and amber/red action states. |
| PIPELINE BOARD | Five stage cards: Extract -> Clean -> Chunk -> Embed -> Index, with per-stage counts and live status.     |
| DOCUMENT TABLE | Name, format, source, version, pages, status, failure reason, retry, inspect.                             |
| DRAWER         | Chunk preview with exact page/clause, metadata and embedding/index status.                                |

### 2. Copilot Workspace

|                   |                                                                                               |
|-------------------|-----------------------------------------------------------------------------------------------|
| HEADER            | Navy header; domain/twist badge; scope/version filter; provider/model chip.                   |
| QUESTION COMPOSER | Compact textarea; Enter-to-run; Cancel during streaming; clear refusal/error state.           |
| ANSWER PANEL      | Streaming response with citation chips; grounded/refused badge; copy/export action.           |
| EVIDENCE DRAWER   | Exact chunks, document/version, page/clause, relevance channel, fusion score and source link. |
| RUN PROGRESS RAIL | Supervisor + three domain specialists + tools + approval, with live event states.             |

### 3. Approval Gate

|               |                                                                                                      |
|---------------|------------------------------------------------------------------------------------------------------|
| QUEUE ITEM    | Risk badge, requester, age, assigned reviewer, runId, proposed action.                               |
| EVIDENCE PANE | Source snippets and citations supporting the proposed action.                                        |
| ACTION PANE   | Read-only original draft; editable structured form for edit-and-approve; comment box.                |
| ACTIONS       | Approve / Edit-and-Approve / Reject; destructive or consequential actions never run before approval. |

|             |                                                                        |
|-------------|------------------------------------------------------------------------|
| AUDIT STRIP | Actor, timestamp, previous/new state, reason and payload hash/version. |
|-------------|------------------------------------------------------------------------|

#### 4. Trace & Metrics Inspector

|                 |                                                                                                     |
|-----------------|-----------------------------------------------------------------------------------------------------|
| RUN HEADER      | Run ID, correlation ID, status, latency, model, tokens, cost.                                       |
| TIMELINE        | Nested spans for retrieval, supervisor, each specialist, tool calls, approval and provider calls.   |
| RETRIEVAL PANEL | Dense top-k, keyword top-k, fusion, filters, selected chunks and citation mapping.                  |
| COST PANEL      | Prompt/completion/total tokens, estimated/actual cost, provider/model breakdown.                    |
| EVALUATION LINK | Jump from a run to its evaluation case, expected evidence and pass/fail rationale where applicable. |

#### 5. Component tree

AppShell -> Sidebar -> Topbar -> RouteContent. Routes: /dashboard, /corpus, /copilot, /reviews, /runs/:runId, /evaluation, /settings. Reusable components: StatusBadge, PriorityChip, EvidenceBlock, CitationChip, TraceSpan, ApprovalPanel, CostCard, ProgressTimeline.

#### 6. Minimal HTTP API surface

| Verb | Endpoint                        | Purpose                               |
|------|---------------------------------|---------------------------------------|
| POST | /api/auth/login                 | Authenticate user / session.          |
| GET  | /api/me                         | Current user and roles.               |
| POST | /api/documents                  | Upload and create ingestion job.      |
| GET  | /api/documents                  | List documents/status.                |
| GET  | /api/documents/:id              | Document/version/chunk summary.       |
| POST | /api/documents/:id/reingest     | Idempotent re-ingestion.              |
| POST | /api/queries                    | Ask grounded question and create run. |
| GET  | /api/runs/:id/stream            | SSE token/progress stream.            |
| POST | /api/runs/:id/cancel            | Server-side cancellation.             |
| GET  | /api/runs/:id                   | Run trace and summary.                |
| GET  | /api/approvals                  | Reviewer queue.                       |
| POST | /api/approvals/:id/approve      | Approve.                              |
| POST | /api/approvals/:id/edit-approve | Edit and approve.                     |
| POST | /api/approvals/:id/reject       | Reject with reason.                   |
| GET  | /api/evaluation/runs            | Evaluation summary.                   |
| GET  | /healthz                        | Liveness.                             |
| GET  | /readyz                         | Readiness.                            |

#### 7. Suggested database core

| Table             | Key fields                                              | Why                      |
|-------------------|---------------------------------------------------------|--------------------------|
| users             | id, email, role, status, created_at                     | Server-side auth/RBAC.   |
| sessions          | id, user_id, title                                      | Persistent history.      |
| documents         | id, source, name, current_version_id, status            | Corpus identity.         |
| document_versions | id, document_id, version, content_hash, language, pages | Version-aware retrieval. |
| chunks            | id, document_version_id, section, page,                 | Citation source unit.    |

| Table                     | Key fields                                                                        | Why                    |
|---------------------------|-----------------------------------------------------------------------------------|------------------------|
|                           | clause, text, metadata                                                            |                        |
| <b>chunk_embeddings</b>   | chunk_id, model, vector                                                           | pgvector index.        |
| <b>ingestion_jobs</b>     | id, document_version_id, stage, status, error_code                                | Pipeline state.        |
| <b>runs</b>               | id, session_id, correlation_id, status, started_at, ended_at                      | Inspectable execution. |
| <b>run_steps</b>          | id, run_id, step_type, agent, status, started_at, ended_at                        | Trace timeline.        |
| <b>tool_calls</b>         | id, run_step_id, tool, args_hash, outcome                                         | Safe tool audit.       |
| <b>approvals</b>          | id, run_id, status, reviewer_id, original_hash, approved_hash                     | HITL gate.             |
| <b>approval_events</b>    | id, approval_id, actor_id, action, timestamp, payload_hash                        | Audit timeline.        |
| <b>usage_ledger</b>       | id, run_id, provider, model, prompt_tokens, completion_tokens, total_tokens, cost | Cost accounting.       |
| <b>evaluation_cases</b>   | id, question, expected_chunks, expected_answer, category                          | Golden set.            |
| <b>evaluation_results</b> | id, case_id, run_id, retrieval_hit, groundedness, refusal_correct                 | Metrics.               |